Suryansh Aryan

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Education

Kevin T. Krofton Department of Aerospace Engineering, Virginia Tech — Blacksburg, VA, USA
M.S Thesis in Aerospace Engineering, Specialization in Dynamics & Control, CGPA: 3.85/4.00 — *Aug 2023 – May 2026*

Manipal University — Manipal, KN, India
B.Tech in Aerospace Engineering, Minor in Computational Mathematics CGPA: 9.09/10.00 — *July 2019 – July 2023*

Experiences

Lead Graduate Research Assistant — May 2024 – Present
Center for Space Science and Engineering, Virginia Tech — *Blacksburg, VA, USA*

- Spearheaded NSF-funded project development as part of a space engineering team, driving LEO satellite constellation **ground software solutions for near-Earth and deep space communication resiliency**.
- Developed and maintained Virginia Tech's in-house deep space emulator SpaceNet, a **MiniNet-based network topology and mission operations testbed**, outperforming state-of-the-art emulators like Hypatia and NS3.
- Led development of **realistic user traffic models, satellite scheduling scenarios, and automated network workflows**, supporting **fault-tolerant and delay/disruption-tolerant networking (DTN)** studies.
- Contributed to **system-level integration, testing, and debugging of distributed space communication systems**, with publications at SciTech 2025 & SmallSat 2025.

UAV Research Intern — Jan 2023 – July 2023
Computational Intelligence Lab, Indian Institute of Science (IISc) — *Bengaluru, India*

- Built autonomous **system software and simulation pipelines** for UAV-based sensing, integrating real-time data processing and system-level testing.
- Built GNC modules with SPI & CAN-based radars & depth camera drivers for PX4 & K++ flight controllers, enabling **robust system integration** and **operational reliability**.
- Designed ROS/Gazebo-based simulations for end-to-end mission workflows, improving canopy coverage efficiency by 40%.

Guidance-Navigation-Control Intern — May 2022 – Aug 2022
General Aeronautics Pvt Ltd — *Bengaluru, India*

- Led the development of **scalable SITL & HITL framework** enhancing trajectory planning for the company's Sense & Avoid project, saving $1000+ on expensive flight controllers & high-resolution sensors.
- Implemented algorithms for **real-time decision-making** and **fault-tolerant navigation**, reducing testing effort and improving system robustness, saving 3+ months of the control division's efforts.
- Implemented a robust metaheuristic GNC algorithm with APF collision avoidance in SITL, enhancing efficiency by 20%, saving 100+ hours of ground testing.

Projects

Fault Tolerant Lunar C&PNT Architecture (Masters Thesis) — Jan 2025 – Apr 2026

- Developed a semi-deterministic Delay Tolerant Network (DTN) emulator for sustainable distributed and fault-tolerant lunar communication & PNT service studies under uncertain dynamics.
- Built automated communication and navigation workflows using BPv7 protocol and TCP convergence layer for **interoperable space network services**.
- Built GNSS-like navigation infrastructure with least square & EKF solutions for **real-time mission scenarios and distributed system simulations**.

Distributed Fault-tolerant Asteroid Inspection | *MATLAB, Simulink, SPICE toolkit*

- Constructed an integrated Adhoc SMC controller with stochastic MPC controller to incorporate unmatched uncertainties in asteroid dynamics under near-frozen orbit conditions for distributed asteroid inspection problem.
- Built a swarm reconfiguration strategy that generates optimal satellite-target pairs and compute desired reference trajectories for the adaptive controller using collocation-based BVP formulation.

Research Publications & Proceedings

1. **Aryan S.** et. al., LEO Satellite Network Testbed: MiniNet-based Emulation of Network Congestion and Routing Dynamics. Acta Astronautica. (In-Progress)
2. B. Barbour, Kedrowitsch A., Downs J., and Mhadgut D., **Aryan S.**, Black J., Kenyon S. "SpaceNet: A Resource-Efficient Simulation Testbed for LEO Mega-Constellation Networks," in IEEE Access, vol. 14, pp. 28097-28118, 2026, doi: 10.1109/ACCESS.2026.3666138. (IEEE Journal)
3. **Aryan, S.**, Downs J., Kedrowitsch A., Smith J., Houck A. and, Kenyon S. SpaceNet Testbed – Small Satellite Network Emulation of Real-World Scenarios. DigitalCommons@USU - *Small Satellite Conference 2025*, Salt Lake City, Utah. (Poster presentation - 3^{rd} best student poster award)
4. **Aryan, S.** Reference tracking of nonlinear airborne systems using stochastic MPC with disturbance observers and actuator chance constraint optimization. CEAS Aeronaut J (2025). https://doi.org/10.1007/s13272-025-00847-w (Springer Nature journal)
5. Downs, J. & Barbour, B. & Kedrowitsch, A. & Mhadgut, D. & **Aryan, S.** & Kenyon, S. Space Network (SpaceNet) Testbed - Development of a Multi-Functional Testbed for Simulating Space Communication Networks. *AIAA Scitech 2025*. 10.2514/6.2025-2716. (Conference proceedings)
6. **Aryan, S.** & Fitzgerald, R. Four Body Invariant Structures And Chaos Analysis For Jovian Multi-Moon Ballistic Transfers. *AAS Astrodynamics Specialist conference 2024, Broomfield, CO.* (Conference proceedings)
7. More, D., Acharya, S., and **Aryan, S.**, "SRGAN-TQT, an Improved Motion Tracking Technique for UAVs with Super-Resolution Generative Adversarial Network (SRGAN) and Temporal Quad-Tree (TQT)," SAE Technical Paper 2022-26-0021, 2022, https://doi.org/10.4271/2022-26-0021. (Conference proceedings)

Technical Skills

Languages: **Python**, **C/C++**, C#, Julia, Markdown, HTML/CSS
Software Tools: **MATLAB/Simulink** (**certified**), **Mathematica**, **Ansys STK** (**certified**), **Robot Operating System (ROS)**, LabView, GMAT, KSP, FlightGear, QGC, MissionPlanner
DevOps & Systems Tools: **Github/GitLab actions**, **Docker**, DigitalOcean, VS Code, Visual Studio, PyCharm, Jupyter notebook, Google colab
Testing & Integration: Pytest, simulation (SITL /HITL)
Libraries: SPICE Toolkit, NumPy, Astropy, Poliastro, Pandas, Scikit, Matplotlib, SkyField, GEOS API, Tensorflow/Keras, OpenCV
Firmware/Hardware: Ardupilot, PX4, Betaflight, Navio2, Raspberry Pi Series, Jetson Nano, Intel NUC, CompactRIO